



SOFTWARE TESTING AND QUALITY ASSURANCE

AI RESUME ANALYZER

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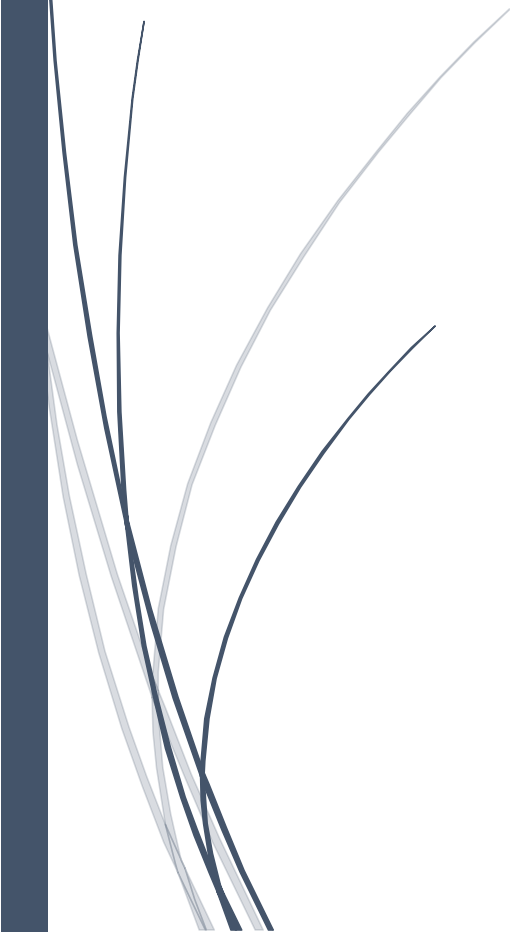
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AI RESUME ANALYZER USING MACHINE LEARNING AND FLASK

ABSTRACT

In today's highly competitive job market, a well-structured resume plays a critical role in securing employment opportunities. Recruiters often receive hundreds of resumes for a single vacancy, making manual screening a challenging and time-consuming task. To address this issue, an intelligent automated system named "AI Resume Analyzer" has been developed.

The AI Resume Analyzer is a web-based application that uses Natural Language Processing (NLP) and Machine Learning techniques to evaluate resumes automatically. Users can upload their resumes in PDF format, after which the system extracts textual information, preprocesses the content, and analyzes various resume components such as education, skills, projects, certifications, and work experience.

Based on the extracted information, the system generates a resume quality score, classifies the resume into predefined categories (Strong, Average, or Weak), and provides personalized suggestions for improvement. The project demonstrates the practical application of Artificial Intelligence in recruitment assistance and career development.

INTRODUCTION

BACKGROUND

The recruitment process has evolved significantly with the growth of technology. Organizations increasingly rely on Applicant Tracking Systems (ATS) and automated screening tools to manage large numbers of job applications. Despite these advancements, many students and fresh graduates still struggle to create effective resumes that satisfy industry standards.

A resume serves as the first impression of a candidate. Missing skills, incomplete project descriptions, poor formatting, or lack of professional experience can significantly reduce a candidate's chances of being shortlisted.

The AI Resume Analyzer aims to bridge this gap by helping candidates assess the quality of their resumes before submitting them to employers.

IMPORTANCE

Many job seekers spend considerable time creating resumes but receive little feedback regarding their effectiveness. Professional resume reviews are often expensive and inaccessible to students.

This project provides:

- Instant resume evaluation
- Automated feedback generation
- Resume quality classification
- Improvement recommendations
- Reduced recruiter workload

PROBLEM STATEMENT

The traditional resume screening process presents several challenges:

- Recruiters spend significant time reviewing resumes manually.
- Human evaluation may introduce bias and inconsistency.
- Students often submit poorly structured resumes.
- Missing critical information reduces hiring chances.
- There is limited access to affordable resume review services.

An automated system capable of evaluating resumes objectively and consistently can significantly improve recruitment efficiency and candidate preparation.

OBJECTIVES

The major objectives of the project are:

PRIMARY OBJECTIVES

- Develop a web-based AI-powered resume analyzer.
- Extract textual information from PDF resumes.
- Apply Machine Learning techniques for evaluation.
- Generate a resume quality score.
- Classify resumes into quality categories.
- Provide intelligent recommendations.

SECONDARY OBJECTIVES

- Improve candidate awareness of resume quality.
- Demonstrate NLP applications in recruitment.
- Reduce manual screening effort.
- Provide a foundation for future ATS integration.

SCOPE

The project focuses on:

FEATURES:

- PDF resume upload
- Text extraction
- Resume preprocessing
- Skill identification
- Resume scoring
- Resume classification
- Feedback generation

SOFTWARE DEVELOPMENT METHODOLOGY

AGILE METHODOLOGY

The Agile development methodology was selected because it supports iterative development, continuous feedback, and flexible requirement changes.

Phase 1: Requirement Gathering

Requirements were collected through:

- Research papers
- Recruiter recommendations
- ATS guidelines
- Online resume standards

Phase 2: Planning

Project modules were identified:

- Frontend Module
- Backend Module
- Resume Parser
- Machine Learning Model
- Feedback Generator

Phase 3: Design

System architecture and workflow diagrams were created.

Phase 4: Development

Modules were developed independently and integrated gradually.

Phase 5: Testing

Unit, integration, and system testing were conducted.

Phase 6: Deployment

The Flask application was deployed locally for demonstration.

Phase 7: Documentation

Complete project documentation was prepared.

PROJECT MANAGEMENT USING TRELLO

Trello was used for task management.

BACKLOG

- Literature review
- Dataset collection
- Requirement analysis

TO DO

- Frontend implementation
- Model training
- Feedback system

IN PROGRESS

- Resume parser
- Classification model

TESTING

- Functional testing
- Performance testing

COMPLETED

- Resume upload module
- AI prediction engine
- User interface

BENEFITS:

- Better collaboration
- Sprint management
- Progress monitoring
- Task prioritization

TOOLS & TECHNOLOGIES

Programming Languages

- Python
- HTML
- CSS
- JavaScript

Framework

- Flask

Machine Learning Libraries

- Scikit-Learn
- NumPy
- Pandas

NLP and Parsing

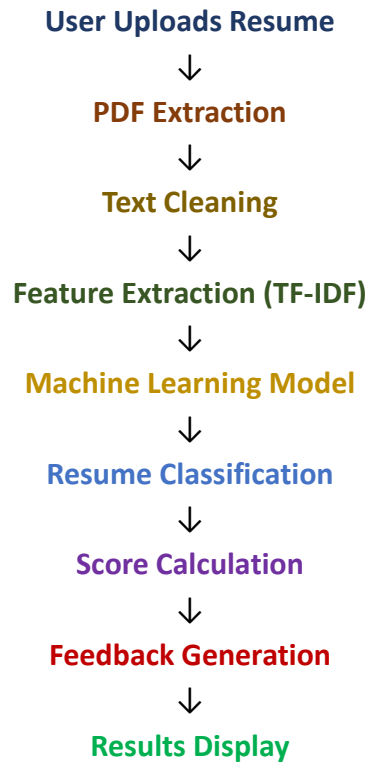
- PDFMiner
- Regular Expressions

Development Tools

- Visual Studio Code
- Git
- GitHub
- Trello

SYSTEM ARCHITECTURE

Workflow:



MACHINE LEARNING IMPLEMENTATION

Dataset Preparation

The training dataset consisted of resumes categorized into:

- Strong Resume
- Average Resume
- Weak Resume

Each resume was manually labeled.

Data Preprocessing

Steps included:

- Lowercase conversion
- Stop-word removal

- Special character removal
- Tokenization

Feature Extraction

TF-IDF Vectorization was used to convert text into numerical features.

Benefits:

- Captures important keywords
- Reduces influence of common words
- Efficient for text classification

Model Selection

Logistic Regression was selected because:

- Fast training
- High interpretability
- Suitable for text classification
- Good performance on sparse data

RESUME SCORING CRITERIA

The system evaluates resumes using weighted criteria.

Section	Weight
Skills	25%
Education	15%
Projects	20%
Experience	20%
Certifications	10%
Contact Information	5%
Resume Structure	5%

Total Score = 100%

Score Interpretation

90–100 = Excellent Resume

75–89 = Strong Resume

60–74 = Average Resume

Below 60 = Weak Resume

SAMPLE RESUME ANALYSIS

Sample Candidate CV

Name: Ali Ahmed

Email: ali.ahmed@gmail.com

Education: BS Computer Science

CGPA: 3.6

Skills:

- Python
- Flask
- Machine Learning
- SQL
- Git

Projects:

1. AI Resume Analyzer
2. Student Management System
3. Face Recognition Attendance System

Certifications:

- Google Data Analytics
- Python for Everybody

Experience:

6-Month Internship at XYZ Software House

SYSTEM EVALUATION

Skills Found

✓ Python

✓ Flask

✓ Machine Learning

✓ SQL

✓ Git

Score: 24/25

Education

✓ Degree Mentioned

✓ CGPA Mentioned

Score: 14/15

Projects

✓ Three Relevant Projects

Score: 18/20

Experience

✓ Internship Experience

Score: 16/20

Certifications

✓ Two Certifications

Score: 8/10

Contact Information

✓ Complete

Score: 5/5

Structure

✓ Proper Formatting

Score: 5/5

Final Result

Total Score:

90/100

Classification:

Strong Resume

Feedback:

The resume demonstrates strong technical skills, relevant projects, certifications, and practical experience. Adding more industry experience and quantifiable project achievements could further improve the profile.

TESTING

Unit Testing

Modules Tested:

- Upload Module
- PDF Parser
- Scoring Engine
- Classification Engine

Integration Testing

Verified communication between:

- Frontend
- Backend
- ML Model

System Testing

Complete application tested successfully.

RESULTS

The developed system successfully:

- Extracts PDF content
- Processes text accurately
- Classifies resumes
- Generates meaningful scores
- Provides useful recommendations

Model achieved satisfactory performance during testing and demonstrated consistent classification results.

ADVANTAGES AND LIMITATIONS

Advantages

- Fast evaluation
- Consistent assessment
- Reduced recruiter effort
- User-friendly interface
- Automated feedback

Limitations

- PDF only
- Dataset-dependent accuracy
- Limited classification categories
- Does not analyze visual design

FUTURE ENHANCEMENTS

Future versions may include:

- ATS Compatibility Score
- Resume vs Job Description Matching
- Skill Gap Analysis
- Deep Learning (BERT)
- GPT-based Feedback System
- Resume Ranking
- Dashboard Analytics
- Cloud Deployment
- Interview Recommendation Engine

CONCLUSION

The AI Resume Analyzer successfully demonstrates the application of Artificial Intelligence and Machine Learning in recruitment support systems. The project automates resume evaluation by extracting information from PDF files, processing textual content, and predicting resume quality using a Logistic Regression classifier. The generated score and personalized feedback help candidates improve their resumes before applying for jobs. This project serves as a practical example of how AI can increase efficiency, reduce manual effort, and enhance decision support in modern recruitment processes.

References

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2. Scikit-Learn Documentation
3. PDFMiner Documentation
4. GitHub Open Source Resume Classification Projects
5. Research Papers on Resume Screening Systems